

Q.1) Display all the files from current directory whose size is greater than n Bytes Where n is accepted from user.

```
#include <stdio.h>

#include <stdlib.h>

#include <dirent.h>

#include <sys/stat.h>


int main() {

    DIR *dir;

    struct dirent *entry;

    struct stat st;

    long n;


    printf("Enter minimum file size in bytes: ");

    scanf("%ld", &n);


    dir = opendir(".");

    if (!dir) {

        perror("opendir");

        return 1;

    }


    printf("Files larger than %ld bytes:\n", n);

    while ((entry = readdir(dir)) != NULL) {

        if (stat(entry->d_name, &st) == -1) {

            perror(entry->d_name);

            continue;

        }


        // Check if it is a regular file and its size

        if (S_ISREG(st.st_mode) && st.st_size > n) {

            printf("%s\t%ld bytes\n", entry->d_name, st.st_size);

        }

    }

}
```

```

}

closedir(dir);

return 0;

}

```

Q.2) Write a C program to find file properties such as inode number, number of hard link, File permissions, File size, File access and modification time and so on of a given file using stat() system call.

```

#include <stdio.h>

#include <stdlib.h>

#include <sys/stat.h>

#include <time.h>

int main(int argc, char *argv[]) {

    if (argc != 2) {

        fprintf(stderr, "Usage: %s <filename>\n", argv[0]);

        return 1;

    }

    struct stat st;

    if (stat(argv[1], &st) == -1) {

        perror("stat");

        return 1;

    }

    printf("File: %s\n", argv[1]);

    printf("Inode number: %ld\n", st.st_ino);

    printf("Number of hard links: %ld\n", st.st_nlink);

    printf("File size: %ld bytes\n", st.st_size);

    printf("File permissions: ");

    printf( (S_ISDIR(st.st_mode)) ? "d" : "-");

    printf( (st.st_mode & S_IRUSR) ? "r" : "-");

```

```
printf( (st.st_mode & S_IWUSR) ? "w" : "-");  
printf( (st.st_mode & S_IXUSR) ? "x" : "-");  
printf( (st.st_mode & S_IRGRP) ? "r" : "-");  
printf( (st.st_mode & S_IWGRP) ? "w" : "-");  
printf( (st.st_mode & S_IXGRP) ? "x" : "-");  
printf( (st.st_mode & S_IROTH) ? "r" : "-");  
printf( (st.st_mode & S_IWOTH) ? "w" : "-");  
printf( (st.st_mode & S_IXOTH) ? "x" : "-");  
printf("\n");
```

```
printf("Last access time: %s", ctime(&st.st_atime));  
printf("Last modification time: %s", ctime(&st.st_mtime));  
printf("Last status change time: %s", ctime(&s
```